

# Day -4 : Network Cable Types and Tools Theory

## 1. Difference between Ethernet Cable and Fibre Optic Cable

### 3 Key Differences

#### 1. Transmission Medium & Signal Type

- **Ethernet Cables:** Use **copper wires** inside to transmit data using **electrical signals**.
- **Fiber Optic Cables:** Use thin strands of **glass or plastic fibers** to transmit data as **pulses of light**.

#### 2. Distance Capabilities

- **Ethernet Cables:** Have a strict distance limit (typically **up to 100 meters / 328 feet**). Beyond this length, the electrical signal degrades rapidly and needs repeaters.
- **Fiber Optic Cables:** Can carry data over **tens of kilometers** without significant loss of signal quality, making them ideal for long-distance transmission.

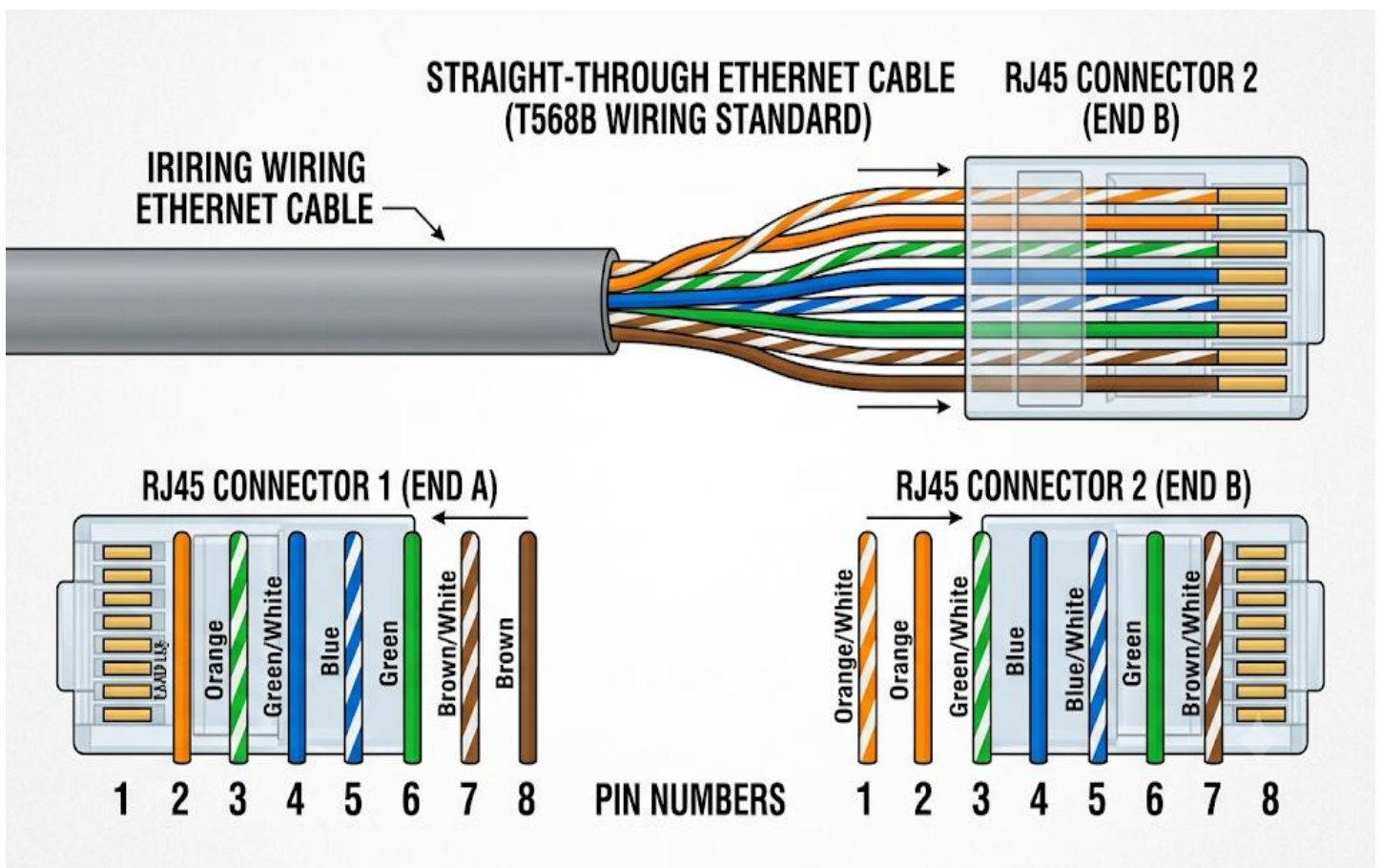
#### 3. Immunity to Interference (EMI)

- **Ethernet Cables:** Susceptible to **Electromagnetic Interference (EMI)** from power lines, heavy

machinery, and lightning, which can slow down or corrupt data.

- **Fiber Optic Cables:** Immune to EMI because light signals are not affected by magnetic fields or electrical noise.

## 2. Image Of Ethernet cable



### 3. How To Use RJ45 Crimper

- First take CAT6 Cable
  - Take RJ45 Connector
  - Take wire stripper tool or scissor
  - Take RJ45 crimping Tool
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- i. Take the CAT 6 cable and cut the from the starting point
  - ii. Then untwist the colour wire inside the CAT 6 cable like White orange orange , white blue , blue , white brown , brown ,white green , green , etc
  - iii. Make the wire straight
  - iv. Then connect the wire to the RJ45 connector
  - v. At the last step crimp the RJ45 connector , Squeeze the tool handles tightly together until it clicks or squeezes fully.

## 4. Gaming Café Setup

- For a 10-PC LAN gaming setup, I would choose **Ethernet cables (CAT6)** paired with an **RJ45 crimper**. Ethernet cables are significantly cheaper, more flexible, and far easier to install across short indoor distances than delicate fibre optic cables.
- CAT6 easily delivers gigabit speeds with virtually zero latency, which is more than enough for gaming